

Cloud Deployment Project Using Google Cloud Storage (GCP)

Student Information

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Program: Computer Engineering

Course: Cloud Computing

Project Title: Static Website Deployment Using Google Cloud Storage (GCP)

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1. Introduction

Cloud computing enables users to store data, host applications, and access computing resources over the internet. Google Cloud Platform (GCP) provides a variety of cloud services, including Cloud Storage, which can be used to host static websites.

This project demonstrates the deployment of a simple static website using Google Cloud Storage. The website is publicly accessible through a cloud-hosted URL and showcases the basic concepts of cloud deployment and web hosting.

2. Project Objective

The primary objectives of this project are:

- To understand cloud storage services.
- To create and configure a Google Cloud Storage bucket.
- To deploy a static website on the cloud.
- To make the website publicly accessible.
- To gain practical experience with cloud deployment technologies.

3. Tools and Technologies Used

Tool/Technology	Purpose
Google Cloud Platform (GCP)	Cloud Service Provider
Google Cloud Storage	Website Hosting
HTML5	Website Development
Web Browser	Testing and Access

4. Project Architecture

The deployment architecture is simple and consists of three components:

User Browser



Google Cloud Storage Bucket



Static Website (index.html)

When a user enters the website URL, the browser sends a request to the Google Cloud Storage bucket, which serves the HTML page to the user.

5. Implementation Steps

Step 1: Create a Static Website

A simple HTML webpage was created using HTML5.

File Name:

index.html

The webpage contains:

- Project title
- Deployment success message
- Developer information
- Basic styling and formatting

Step 2: Create a Cloud Storage Bucket

A new storage bucket was created in Google Cloud Storage with the following configuration:

- Unique Bucket Name
- Storage Class: Standard
- Location Type: Multi-Region
- Access Control: Uniform
- Encryption: Google Managed

Step 3: Upload Website Files

The index.html file was uploaded to the bucket through the Google Cloud Console.

Step 4: Configure Public Access

To allow public access to the website:

Principal:
allUsers

Role:
Storage Object Viewer

This configuration enables users on the internet to access the website.

Step 5: Access the Website

After deployment, the website became accessible through a public URL generated by Google Cloud Storage.

Website URL:

<https://storage.googleapis.com/cloud-deployment-fatima-2026-001/index.html>

6. Website Source Code and Screen Shots

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Cloud Deployment Project</title>
```

```
<style>
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    text-align: center;
```

```
    background: #f4f6f8;
```

```
    margin-top: 80px;
```

```
  }
```

```
  .card {
```

```
background: white;
width: 60%;
margin: auto;
padding: 30px;
border-radius: 10px;
box-shadow: 0px 0px 15px rgba(0,0,0,0.1);
}
```

```
h1 {
  color: #2c3e50;
}
```

```
p {
  font-size: 16px;
  color: #444;
}
```

```
.footer {
  margin-top: 20px;
  font-weight: bold;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

<h1> 🚀 Cloud Deployment Successful</h1>

<p>This website is hosted using Google Cloud Storage (GCP).</p>

<p>This project demonstrates static website deployment using cloud infrastructure.</p>

<hr>

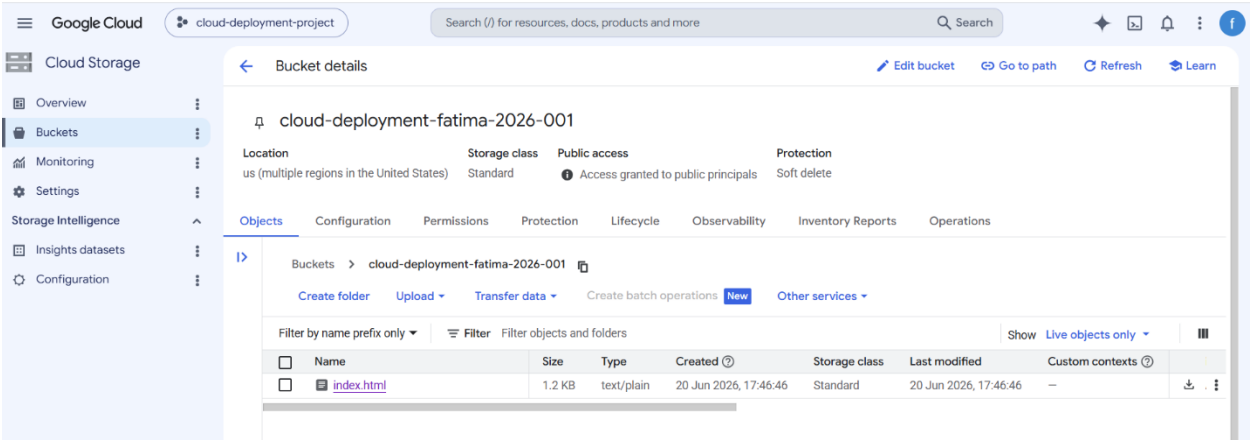
<p class="footer">Developer: Fatima Hameed</p>

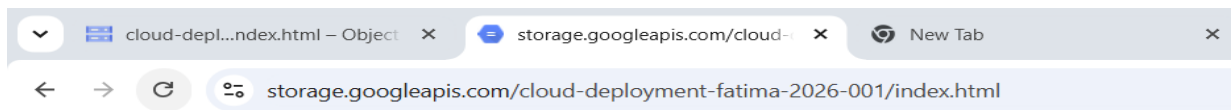
<p class="footer">Course: Cloud Computing Assignment</p>

</div>

</body>

</html>





```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Cloud Deployment Project</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      background: #f4f6f8;
      margin-top: 80px;
    }

    .card {
      background: white;
      width: 60%;
      margin: auto;
      padding: 30px;
      border-radius: 10px;
      box-shadow: 0px 0px 15px rgba(0,0,0,0.1);
    }

    h1 {
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    p {
      font-size: 16px;
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    .footer {
      margin-top: 20px;
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    }
  </style>
</head>
<body>
<div class="card">

  <h1>ðŸŒ€ Cloud Deployment Successful</h1>

  <p>This website is hosted using <b>Google Cloud Storage (GCP)</b>.</p>

  <p>This project demonstrates static website deployment using cloud infrastructure.</p>

  <hr>

  <p>This project demonstrates static website deployment using cloud infrastructure.</p>

  <hr>

  <p class="footer">Developer: Fatima Hameed</p>
  <p class="footer">Course: Cloud Computing Assignment</p>

</div>

</body>
</html>
```

7. Results

The website was successfully deployed on Google Cloud Storage and made publicly accessible through a cloud-hosted URL.

Achievements:

- Successfully created a cloud storage bucket.
- Uploaded website files to the cloud.
- Configured public access permissions.
- Accessed the website through a public URL.
- Verified successful cloud deployment.

8. Screenshots

Insert the following screenshots:

Screenshot 1: Google Cloud Storage Bucket Creation

Screenshot 2: Uploaded index.html File

Screenshot 3: Bucket Permissions Showing allUsers Access

Screenshot 4: Live Website Running in Browser

9. Challenges Faced

During deployment, the following issues were encountered:

- Incorrect file naming (index.html.txt instead of index.html)
- Missing object errors (NoSuchKey)
- Bucket configuration issues
- Public access permission configuration

These issues were resolved by correcting file names, recreating the bucket, and configuring proper access permissions.

10. Conclusion

This project successfully demonstrated the deployment of a static website using Google Cloud Storage. The project provided practical experience with cloud infrastructure, storage

management, website hosting, and access control. Through this implementation, a better understanding of cloud deployment processes and web hosting concepts was achieved.

The project confirms that Google Cloud Storage can be effectively used as a reliable and scalable platform for hosting static websites.

References

1. Google Cloud Platform Documentation
2. Google Cloud Storage Documentation
3. HTML5 Official Documentation
4. Cloud Computing Fundamentals